

Project Title

Potato Leaf Disease Detection using Image Classification

Project Description

This project focuses on building an image classification model to detect and classify potato plant diseases. The dataset consists of high-resolution images of potato leaves categorized into three classes: Early Blight, Late Blight, and Healthy leaves. The goal is to develop a robust deep learning model that can accurately identify plant diseases and support agricultural diagnostics.

Objectives

- Understand image classification using deep learning
- Perform data preprocessing and augmentation
- Train and evaluate a CNN-based model

Dataset Details

- Classes:
 - Early Blight
 - Late Blight
 - Healthy
- Data Type: Image dataset (high-resolution leaf images)

Code to load dataset

<https://www.kaggle.com/datasets/hafiznouman786/potato-plant-diseases-data>

```
import kagglehub
```

```
# Download latest version
```

```
path = kagglehub.dataset_download("hafiznouman786/potato-plant-diseases-data")
```

```
print("Path to dataset files:", path)
```

Project Tasks

1. Data Understanding

- Load and explore the dataset
- Visualize sample images from each class

2. Data Preprocessing

- Resize images
- Normalize pixel values
- Apply data augmentation (rotation, flipping, etc.)

3. Model Building

- Build a CNN model
- Define architecture, loss function, and optimizer

4. Model Training

- Train the model on training data
- Validate using validation dataset
- Test the overall model performance

5. Model Evaluation

- Evaluate using metrics:
 - Accuracy
 - Confusion Matrix
 - Classification Report

6. Record a video for 15 mins